



LeetCode 9: Palindrome Number

1. Problem Title & Link

Title: LeetCode 9: Palindrome Number

Link: <https://leetcode.com/problems/palindrome-number/>

2. Problem Statement (Short Summary)

Given an integer x , determine whether it is a palindrome.

A palindrome number reads the same forward and backward.

Return:

- true if it is a palindrome
- false otherwise

Examples:

121 → Palindrome

1221 → Palindrome

12321 → Palindrome

123 → Not Palindrome

120 → Not Palindrome

3. Examples (Input → Output)

Example 1

Input:

$x = 121$

Output:

true

Explanation:

Forward = 121

Backward = 121

Example 2

Input:

$x = -121$

Output:

false

Explanation:

Forward = -121



Backward = 121-

Not the same.

Example 3

Input:

x = 10

Output:

false

Explanation:

Forward = 10

Backward = 01

Not equal.

Example 4

Input:

x = 0

Output:

true

Explanation:

0 reads the same both ways.

4. Constraints

$-2^{31} \leq x \leq 2^{31} - 1$

Follow-Up:

Can you solve it without converting the integer to a string?

5. Core Concept (Pattern / Topic)

Mathematics / Number Manipulation

Secondary Topics:

- Reverse Number
- Digit Extraction
- Modulo Operator

6. Thought Process (Step-by-Step Explanation)

Brute Force Idea

Convert the number into a string.



Example:

121

Convert:

"121"

Reverse:

"121"

Compare both.

If equal:

Palindrome

Else:

Not Palindrome

Simple but uses extra space.

Optimized Thinking

Instead of converting to string:

Reverse the number mathematically.

Example:

$x = 121$

Build reversed number:

1

12

121

Finally compare:

$\text{original} == \text{reversed}$

If same:

Palindrome

Important Observation

Negative numbers can never be palindrome.

Example:

-121

Reverse:

121-

Invalid.

Therefore:

$x < 0 \rightarrow \text{false}$



Another Observation

Numbers ending with 0 cannot be palindrome.

Example:

10

Reverse:

01

Different.

Exception:

0

is palindrome.

7. Visual / Intuition Diagram (ASCII Diagram)

Example:

`x = 121`

Digit extraction:

`121 % 10 = 1`

`reverse = 1`

`12 % 10 = 2`

`reverse = 12`

`1 % 10 = 1`

`reverse = 121`

Final:

`Original = 121`

`Reverse = 121`

Match ✓

Reverse Building Diagram

`reverse = 0`

Digit = 1

`reverse = 0 * 10 + 1`
`= 1`

Digit = 2

`reverse = 1 * 10 + 2`



```
        = 12

Digit = 1

reverse = 12 * 10 + 1
        = 121
```

8. Pseudocode (Language Independent)

```
if x < 0
    return false

original = x
reverse = 0

while x > 0

    digit = x % 10

    reverse = reverse * 10 + digit

    x = x / 10

return original == reverse
```

9. Code Implementation

Python

```
class Solution:
    def isPalindrome(self, x: int) -> bool:

        if x < 0:
            return False

        original = x
        reverse = 0

        while x > 0:

            digit = x % 10
            reverse = reverse * 10 + digit
            x //= 10
```



```
return original == reverse
```

Java

```
class Solution {  
  
    public boolean isPalindrome(int x) {  
  
        if(x < 0)  
            return false;  
  
        int original = x;  
        int reverse = 0;  
  
        while(x > 0) {  
  
            int digit = x % 10;  
  
            reverse = reverse * 10 + digit;  
  
            x /= 10;  
        }  
  
        return original == reverse;  
    }  
}
```

10. Time & Space Complexity

Time Complexity

$O(\log_{10} n)$

Reason:

One iteration per digit.

Example:

121 → 3 digits → 3 iterations

Space Complexity

$O(1)$

Reason:

Only a few variables are used.



11. Common Mistakes / Edge Cases

Mistake 1

Ignoring negative numbers.

Example:

-121

Expected:

false

Mistake 2

Using floating-point division.

Wrong:

$x = x / 10$

Correct:

$x //= 10$

Mistake 3

Modifying original number without storing it.

Wrong:

while $x > 0$:

After loop:

$x = 0$

Cannot compare.

Need:

original = x

Edge Cases

Input	Output
0	TRUE
1	TRUE
11	TRUE
10	FALSE
-121	FALSE

12. Detailed Dry Run (Step-by-Step Table)

Input:

$x = 121$



Initial:

original = 121

reverse = 0

x	digit=x%10	reverse
121	1	1
12	2	12
1	1	121
0	-	Stop

Comparison:

original = 121

reverse = 121

Result:

true

Another Dry Run

Input:

x = 123

x	digit	reverse
123	3	3
12	2	32
1	1	321

Compare:

123 != 321

Output:

false

13. Common Use Cases (Real-Life / Interview)

Numeric Validation

Special IDs

Ticket numbers

Codes

Pattern Recognition

Symmetric data detection



Digital Signal Processing

Checking mirrored patterns

Interview Purpose

Tests:

Modulo

Division

Digit extraction

Loop control

14. Common Traps (Important!)

Trap 1

Converting to string immediately.

Interviewers often ask:

Solve without string conversion.

Trap 2

Not handling:

$x < 0$

Trap 3

Not storing original number.

original = x

is mandatory.

Trap 4

Integer Overflow (Advanced)

For very large values:

$\text{reverse} * 10 + \text{digit}$

may overflow in some languages.

Java constraints here are safe, but interviewers may ask about it.

15. Builds To (Related LeetCode Problems)

Progression:

LC 9 - Palindrome Number

↓

LC 7 - Reverse Integer



LC 66 - Plus One



LC 989 - Add to Array Form



LC 415 - Add Strings

These improve digit manipulation skills.

16. Alternate Approaches + Comparison

Approach	Time	Space	Interview Rating
String Conversion	$O(n)$	$O(n)$	Easy
Full Reverse Number	$O(\log n)$	$O(1)$	Good
Reverse Half Only	$O(\log n)$	$O(1)$	Best

Approach 1: String Conversion

```
return str(x) == str(x)[::-1]
```

Pros:

Very short

Easy

Cons:

Extra space used

Approach 2: Full Reverse Number

(Current Solution)

Reverse complete number

Compare

Pros:

No string conversion

$O(1)$ space

Approach 3: Reverse Half (Optimal)

Instead of reversing entire number:

12321

Left = 12

Right = 12



Reverse only half digits.

Benefits:

Avoids overflow

Less work

This is the official optimal solution.

17. Why This Solution Works (Short Intuition)

Reversing the digits reconstructs the number in reverse order.

If the reversed number equals the original number, the digits are identical from both directions, making it a palindrome.

18. Variations / Follow-Up Questions

Follow-Up 1

Check palindrome for a string.

Example:

"madam"

Output:

true

Follow-Up 2

Ignore spaces and punctuation.

Example:

"A man, a plan, a canal: Panama"

Output:

true

Related: Cracking the Coding Interview interview-style problem.

Follow-Up 3

Find the largest palindrome less than N.

Example:

N = 200

Output:

191

Follow-Up 4

Count palindrome numbers in a range.

Example:



1 → 1000

How many palindrome numbers exist?

Interview Takeaway

For numeric palindrome questions, immediately think:

$\text{digit} = n \% 10$

$\text{reverse} = \text{reverse} * 10 + \text{digit}$

$n = n / 10$

This digit extraction pattern appears repeatedly in coding interviews and competitive programming.